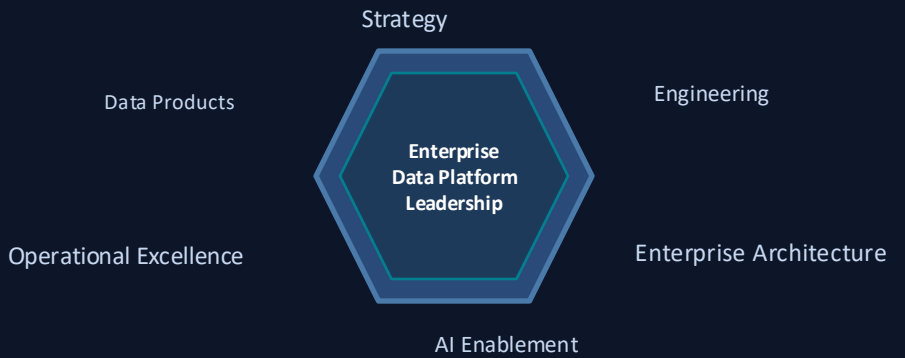


# Jeff Welcome

## Executive Technical Portfolio

Enterprise Data Platforms • Data Engineering • Enterprise Architecture • DataOps Leadership

*A leadership portfolio showcasing enterprise-scale platform architecture, engineering operating models, modernization strategy, and business outcomes.*



**28+**

Building Enterprise Data Capabilities

**1.2 PB**

Protected Enterprise Data Footprint

**100+**

Engineers & Analysts Enabled

**7,000+**

Daily Automated Workloads

Enterprise Platform Strategy • Architecture Modernization • Engineering Leadership • Operational Excellence

# Building at the edge of industry maturity

Three decades of turning emerging technology into production capability — before the playbook existed

1998

## eLocal.com

First technical hire · Data Integration Manager

### Multi-channel data platform

#### Architecture flow

External data providers

- ↓ Canonical geographic model  
city / state / ZIP → lat / lon
- ↓ Web · Mobile · GPS · ITV

#### Key proof points

- 75,000 cities · multi-channel geographic distribution
- Early POC: address → lat/lon for location-aware consumption
- Partnered with AT&T Broadband, Motorola

2002

## Coldwater Creek

Principal Data Warehouse Developer

### Enterprise data at uncommon scale

#### Architecture flow

- SQL Server 2000 (not Oracle · not Teradata)
- ↓ Processing server
- ↓ Metadata-driven table swap
- ↓ Reporting server · near-instant cutover

#### Key proof points

- 26TB · billion-row dimensional models
- 1TB+ nightly · near-instant reporting cutover
- Metadata-driven DTS/SSIS · automated partition mgmt

2011 → 2017 → 2024

## Columbia Sportswear

Lead Engineer → Manager → Sr. Manager

### Repeated platform reinvention

#### 2011 — SAP as an Integration Layer

SAP → BW / OpenHub → Teradata EDW

Extended BW beyond reporting into enterprise data integration



#### 2016–17 — Cloud Modernization

Cloud POCs → Azure / Databricks production  
Production pre-Ignite · ADF V2 + Azure Databricks announced at Ignite



#### 2024 — Federated Ownership at Scale

Lakehouse → federated data mesh  
100+ engineers & analysts enabled

- 1.2PB · 7,000+ workloads · 3,000+ notebooks
- 14 person global org · \$700K+ recurring annual cloud savings

# ENTERPRISE DATA & ANALYTICS PLATFORM ARCHITECTURE

## SOURCE SYSTEMS

SAP · ERP · POS · eCommerce · Vendors · Third-Party Integrations

## METADATA-DRIVEN INGESTION (Azure Data Factory)

Generic Pipelines · Parameterized Orchestration · Schema-Driven Metadata · CI/CD Deployment

## ADLS (Azure Data Lake Storage)

### IMMUTABLE LANDING / RUSTZONE

Raw Immutable Storage · Compliance / Audit-Ready · Zero-Mutation Ingestion

## DATABRICKS LAKEHOUSE — Bronze → Silver → Gold

Medallion Architecture: Raw Delta | Cleaned & Conformed | Curated Business-Ready

### Enterprise Data Products

#### Curated Enterprise Data

Facts · Dimensions · Conformed Data

#### Analytics & Semantic Products

Enterprise KPIs · Semantic Models · Business Ready Datasets

#### Reusable Data Products

Shared Domain Assets · Governed Enterprise Products

### ML / Data Science

#### ML Development

Feature Store · Experimentation · Pipelines

#### Model Lifecycle

Model Registry · Deployment · CI/CD

#### Production Enablement

Orchestration · Scalable APIs · Operational Integration

### Data Delivery & Exchange

#### Operational Data Products

Batch · Micro-Batch · Real-Time · APIs

#### Application & Partner Integration

Internal Systems · External Platforms

#### Governed Data Sharing

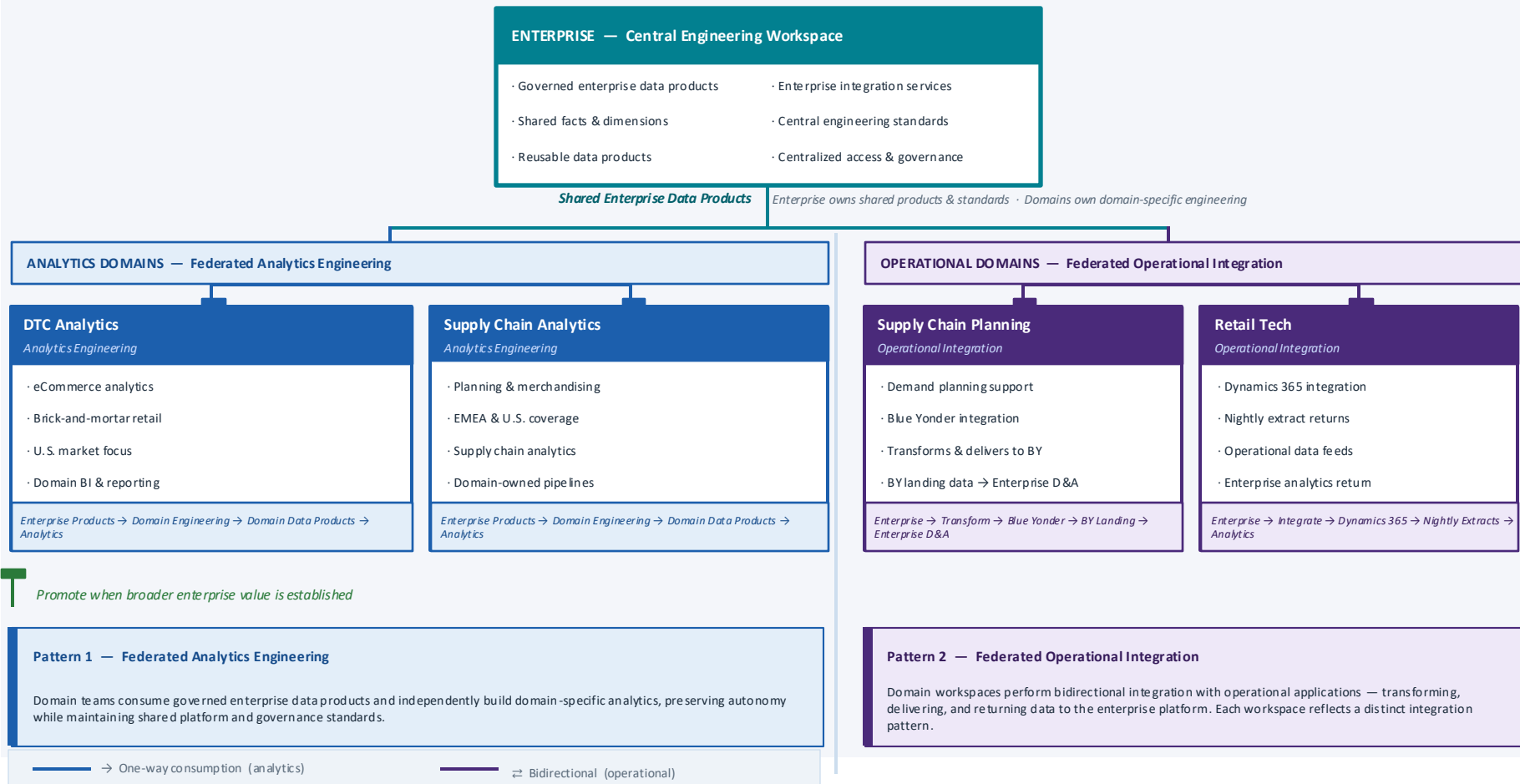
Delta Sharing · External Data Exchange

## DELIVERY — Power BI · APIs · Operational Apps

Enterprise Dashboards · Embedded Analytics · Programmatic Consumption · Self-Service Workspaces

# Federated Data Platform

Five-workspace operating model balancing centralized enterprise data products with domain autonomy for analytics and operational integration



# Enterprise Data Platform Operating Model

*Enabling Enterprise Scale Through Federated Engineering*

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**Platform  
Engineering**

**Enterprise Data  
Engineering**

**Domain  
Engineering**

**Promotion &  
Governance**

Centralized Platform · Federated Ownership · Governed by Design

The Columbia Sportswear Enterprise Data Platform balances centralized platform engineering with domain-level innovation — shared services and governance provide consistency while enabling teams to rapidly develop analytical solutions.

# Engineering Operating Model

Platform Engineering · Enterprise Data Engineering · Domain Engineering & Analytics

## Platform Engineering

### Mission

Deliver and operate the enterprise data platform as a secure, scalable, and reliable product.

### Responsibilities

- Azure platform architecture
- Databricks platform administration
- Unity Catalog governance
- Enterprise networking & security
- Platform observability
- Infrastructure as Code
- CI/CD frameworks
- Disaster Recovery
- Authentication & access management
- Engineering standards
- AI Enablement
- ML-Ready Platform

### Deliverables

· Enterprise Azure Platform · Databricks Platform · Unity Catalog · CI/CD Frameworks · IaC · Monitoring & Observability · Platform Roadmap · Security Standards

### Primary Customers

· Enterprise Data Engineering · Domain Engineering Teams · Data Science & ML Engineering · Enterprise IT · Platform Operations

## Enterprise Data Engineering

### Mission

Develop and operate enterprise data products through standardized integration of enterprise applications with the Lakehouse, enabling trusted bidirectional data movement while providing reusable data products for analytics, reporting, machine learning, operational business processes, and business domains.

### Responsibilities

#### Enterprise Integration Services

- Enterprise application integration
- Metadata-driven ingestion pipelines
- Source system onboarding
- Bidirectional data movement

#### Enterprise Data Products

- Medallion architecture (Bronze / Silver / Gold)
- Enterprise data products
- Shared semantic models
- Operational APIs

#### Engineering

- Metadata-driven orchestration
- Data quality & observability
- Production operations
- Reusable engineering frameworks

### Deliverables

Enterprise Data Products · Metadata-driven ingestion · Enterprise Facts & Dimensions · Curated Datasets · Integration Patterns · Trusted Data Assets · Shared Semantic Models

### Primary Customers

· Business Domains · Domain Engineering and Analytics · Enterprise IT Application Teams · Data Science · Executive Analytics

## Domain Engineering & Analytics

### Mission

Enable business organizations to rapidly develop domain-specific data products, analytical datasets, operational reporting, and business intelligence using enterprise data products as foundational building blocks.

### Responsibilities

- Domain notebooks & datasets
- Power BI reports
- Analytical experimentation
- Business transformations
- Prototype data products
- Self-service analytics
- Department-owned pipelines

### Deliverables

· Domain Data Products · Department Data Pipelines · PowerBI Semantic Models · Business Dashboards · Operational Reporting · Analytical Datasets · Department KPIs · Enterprise Promotion Candidates

### Primary Customers

· Direct To Consumer · Supply Chain · Finance · Sales · Marketing · Ecomm

# Enterprise Promotion & Governance

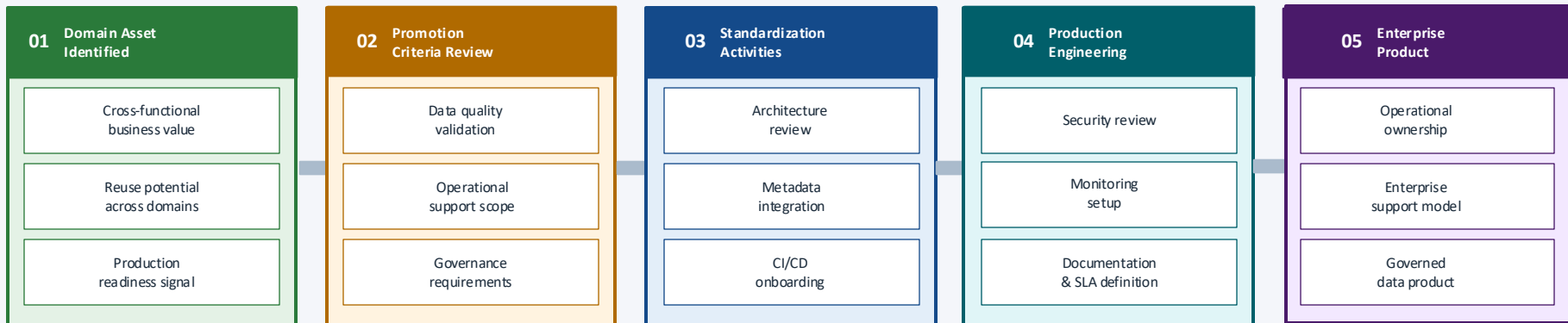
The transition process that evolves successful domain assets into enterprise-supported data products

**Mission** — Evaluate, standardize, and operationalize successful domain assets into enterprise-supported data products.

**Customer:** Enterprise Platform

**Input:** Domain Asset

**Output:** Enterprise Capability



**Outcome** — Business innovation becomes enterprise capability.

Promotion is activated when a domain asset meets these criteria:

## Start in Domain

Teams experiment freely without governance overhead during early-stage work.

## Promote on Value

Assets graduate when they demonstrate cross-functional business value.

## Standardize on Entry

Governance, CI/CD, metadata, and SLAs applied at promotion — not before.

## Enterprise Ownership

Promoted products are owned, operated, and supported by platform teams.

# Operating Principles

*The design philosophy behind the Enterprise Data Platform*

## 01 Platform as a Product

Engineering teams consume shared platform capabilities rather than building infrastructure independently. The platform is designed, operated, and evolved like a product.

## 02 Federated Ownership

Business domains own business logic while platform teams own enterprise capabilities. This reduces bottlenecks and enables parallel delivery at scale.

## 03 Reusable Engineering

Metadata-driven frameworks and common components minimize duplication and accelerate delivery. Build once, reuse everywhere.

## 04 Governed by Design

Enterprise governance is applied when assets become strategic enterprise products — not during early experimentation. Governance enables without blocking.

## 05 Automation First

Standardized CI/CD, Infrastructure as Code, automated deployments, and metadata-driven orchestration reduce manual effort and improve reliability at scale.